Ad-hoc Networking

Most of us have heard of the term ad-hoc networking, but have never quite dwelt on understanding its potential in making our lives much easier

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Literally meaning "for this purpose", this is a network intended for closed functioning. A typical ad-hoc network involves interconnecting devices. It fits into the typical scenario—a desktop, the home PC, along with a couple of laptops—all connected together sharing resources such as storage area effectively, and most importantly, a common Internet connection.

Connecting Your Home Without A Router

Connecting your home with the standard configuration is quite simple. There are three basic steps involved in this. First you need to ensure that all the systems are WiFi compatible (buy a WLAN card for your desktop, if you need to). After you have cleared this hurdle, you need to configure it as a computer-to-computer (ad-hoc) network. Lastly, comes the Internet Connection Sharing (ICS) part.

Getting your computer WiFi-enabled is really simple. You need to plugin a Wireless network card, which is available in two forms—one a USB kind which you just plug and play, and the other which you need to fix into your PCI slot on your motherboard. Most laptops are WiFi enabled, so no work needed there.

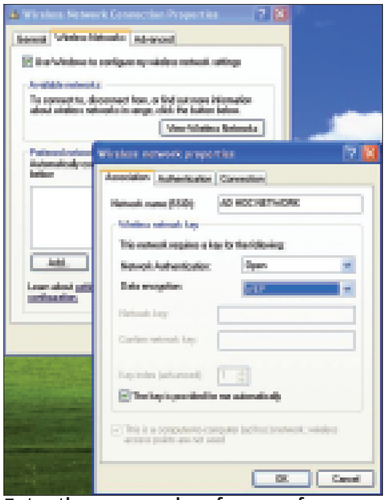
In order to proceed with setting up the network, you need to configure one computer as the host and others as the client. We will start with the host computer. Click on the icon in your system tray to view available wireless connections. You may see some wireless networks from your neighbourhood, in the list. Do not select any yet.

Go to `Start > Control Panel > Network Connections`. Right-click on your wireless network adapter and click on `Properties`. Select the `Wireless Network` Tab.

Click on `Advanced` on the bottom right. Select the option for `Computer-to-computer (ad-hoc) networks only`.

Our next step is to christen our ad-hoc network so that all users can log on to it. To do this, click on `Add` in the same dialog box. You need to enter a name for your ad-hoc network—called a Service Set Identifier (SSID) which is nothing but a name for your network.

To prevent abuse of your ad-hoc network, it is always wise to use the network authentication and data encryption features. Nevertheless, in this scenario, you are still safe even of you leave `Network Authentication` as `Open`—this setting has more relevance if you're setting up a more permanent network.



Enter the name and preferences for your ad-hoc network

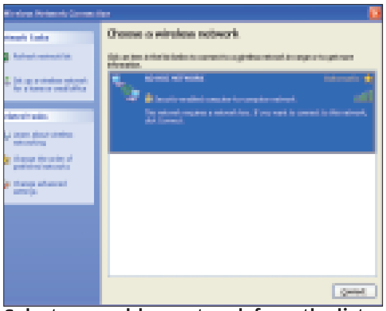
Once you enter in these details, click `OK` and you are done with the host computer.

Configuring The Client

The way to configure the client is in not too different from the way you configured the host. Follow the same procedure, enter the same SSID, and remember to select the option for `Computer-to-computer ad-hoc networks only`.

Once you are done with these steps, you are ready to connect. Now click on the icon for wireless networks. Refresh the network list on the laptops / computers that will act as clients on the ad-hoc network (`AD-HOC NETWORK` in our case).

You will see your ad hoc network appearing in the list of available networks. Select this network and click on `Connect`. It is worth noting that the systems that are part of the ad hoc network should be connected simultaneously to the network in order to make it a more convenient procedure.



Select your ad-hoc network from the list

Once your connection is established, Windows will will indicate the signal strength and quality. Now all you have to do is start sharing! ■

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